Sabina Bimali

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Summary

I am a motivated Ph.D. student in Mechanical Engineering at Purdue University focused on advancing the reliability assessment and metrology of thermal interface materials (TIMs). My research emphasizes the development of experimental techniques to enable reliable thermal management in electronics packaging applications. Actively seeking a **thermal intern** role for Summer 2027.

Relevant Highlights

Industrial Experience: Worked as a summer Thermal Intern at Qualcomm with two projects
(1) Performance testing of mobile devices and
(2) Analytical/Numerical modeling for TIM1.5 selection for electronics packaging
Research Experience: 2 years of research on package thermal behavior/ metrology development for TIMs degradation tests and, 2 years of experience in design, development and reliability testing of flexible hybrid printed electronics.
Publications: IEEE ITherm, ECTC, ASME Interpack and JEP.
Awards/Scholarships: Best E-Poster Award at IEEE ITherm 2026 Conference, Winner of EPS Graduate Student Grant 2026, Purdue Graduate Student Government (PGSG) Travel Award 2026, Best Poster Runner-up Award at ITherm 2024 Conference, IPC (Electronics Foundation) Student Member Scholarships 2023, 2024, and 2025.

Education

Purdue University, West Lafayette, Indiana ***Aug'24 – Present***
PhD in Mechanical Engineering ***3.68/4.0***

Purdue University, West Lafayette, Indiana ***Aug'24 – May'26***
Masters of Science, Mechanical Engineering ***3.68/4.0***

Tribhuvan University, Institute of Engineering (IOE), Dharan, Nepal ***Nov'14 – Dec'18***
Bachelor in Mechanical Engineering ***79.57/100***

Skills

Experimental Skills: Cross plane thermal conductivity, reliability test of TIM, thermal performance testing of mobile devices, Flexible Printed Electronics and its reliability, Rheometer, Run various benchmarks to evaluate the thermal performance of mobile phones
Programming Languages: Python, LaTex, C, Arduino IDE
Softwares & Tools: Microsoft Office, MATLAB, SolidWorks, CATIA, Autodesk Inventor, ANSYS Fluent, COMSOL, ImageJ
Data Analysis Tools: R Programming, SAS

Research Experience

Methodology to Assess the Impact of Surface Flatness and Warpage on TIM Degradation ***Aug'24 – Present***
Supervised by: Dr Amy Marconnet, Funded by: Cooling Technologies Research Center (CTRC) *Purdue University*

- Design and development of a test structure for the Thermal Interface Material (TIMs) degradation test under dynamic warpage.
- Inducing and controlling the warpage thermally/mechanically, and measuring the warpage with a single point laser sensor as well as 2D surface plot with optical .
- Result shows that the pump-out accelerates with higher temperature and warpage.
- Ongoing work: Rheology of TIMs, to connect the degradation with the material properties.

CMOS-Compatible Aerogels for Extreme Thermal Isolation ***Aug'24 – Aug'25***
Supervised by: Dr Amy Marconnet, Funded by: DARPA *Purdue University*

- Develop and demonstrate ultra-low thermal conductivity aerogel materials that can be integrated into stack in a fashion compatible with thermal isolation of adjacent functional blocks.
- Mechanical characterization of Aerogel, such as the coefficient of thermal expansion of the fabricated aerogels.

Biodegradable Substrates Low-Temperature Cure for Sustainable Additively Printed Electronics ***Aug'23 – Jul'24***
Supervised by: Dr Pradeep Lall, Funded by: CAVE3 *Auburn University*

- Design, development, and reliability testing of flexible printed hybrid electronic systems developed with biodegradable substrate, water-based solvents and low-temperature processing.
- Performed electrical and mechanical characterization of printed circuits.

Sustainable Additively Printed Electronics through Water-Solvent Inks ***Aug'22 – Jul'23***
Supervised by: Dr Pradeep Lall, Funded by: CAVE3 *Auburn University*

- Design, development, and reliability testing of flexible printed hybrid electronic systems developed with water-based solvents and low temperature interconnects.

- Performed electrical and mechanical characterization of printed electronic circuits and benchmarked results with standard rigid board circuits as well as the simulated results.
- Results showed a big potential of the sustainable flexible printed circuit boards with comparable performance to the traditional boards.

Industrial Experience

Qualcomm Incorporated ***June'26 – Sept'26***

Thermal Intern *San Diego, California*

- Conducted performance benchmarking and comparative analysis of commercial mobile devices using industry-standard benchmarks to evaluate the system performance.
- Developed Analytical/Numerical model of electronic packages with and without heat spreader lids, analyzing temperature distribution, thermal resistance, and operating conditions under which lidless configurations outperform lidded designs.

G.R.I.T. Engineering Pvt. Ltd ***Sept'18 – Mar'21***

Mechanical Design Engineer *Lalitpur,Nepal*

- 2D/3D drawing/modelling and simulation of mechanical systems and components.
- Collaborated closely with manufacturing teams, providing on-site technical supervision and ensuring accurate implementation of machine layouts.
- Contributed to diverse industrial projects, including plastic recycling machinery, hydraulic molding systems, material handling equipment, marine structures, and custom mechanical systems for commercial and media applications.

CG Industrial Park ***Dec'17 – Jan'18***

Mechanical Intern *Nawalparashi, Nepal*

- Monitoring and recording the daily maintenance tasks for all Noodles making machines owned by the company.
- Compiling weekly repair and maintenance reports for the maintenance manager.

Teaching Experience

Tribhuvan University, Himalaya College of Engineering ***April'21 – July'22***

Assistant Lecturer (Department of Civil Engineering) *Lalitpur, Nepal*

- Taught Engineering Drawing I & II for undergraduate engineers in civil and computer engineering departments.
- Conducted Thermodynamics and Heat Transfer laboratories, providing hands-on training in heat engines and refrigeration systems.
- 3D modeling and visualization to improve student understanding of engineering graphics and thermal concepts.

Leadership Experience

Electronics Foundation- Purdue Chapter ***Sept 2026 – Present***

President *Purdue University*

- Plan, organize, and oversee chapter activities and meetings.
- Develop the annual plan/review, delegate responsibilities, ensure leadership succession, and maintain communication with the Electronics Foundation.

Cooling Technologies Research Center (CTRC) ***May 2025 – Present***

Organizer, CTRC Poster Session *Purdue University*

- Organized and executed CTRC Poster Session, ensuring high-quality event delivery.
- Managed logistics, communications, and participant coordination for a seamless attendee experience.

Nepali Society at Purdue (NEPSAP) ***April 2026 – Present***

Marketing/Design Chair *Purdue University*

- Manage NEPSAP's communications, including social media, email, and public relations.
- Coordinate event publicity and outreach to increase engagement and attendance.

Surface Mount Technology Association (SMTA) ***Jan 2024 – Aug 2024***

Vice President of Speaker Series *Auburn University*

- Led speaker series initiatives and coordinated technical events for the SMTA student chapter.
- Oversaw event logistics and communication to ensure strong engagement and execution.

Honors and Awards

EPS Graduate Student Grant 2026: Winner of EPS Graduate Student Grant 2026, awarded to the outstanding graduate student working in the field of Electronics Packaging. **(Aug 2026)**

IEEE ITherm 2026 Best E-Poster Award: Awarded among 50+ presenters, in Mechanics and Reliability Track at IEEE ITherm Conference. **(May 2026)**

Purdue Graduate Student Government (PGSG) Travel Award: Selected for competitive PGSG Travel Award to present research (paper and poster) at 2026 IEEE ITherm Conference. **(April 2026)**

Electronics Foundation (IPC Education) Student Member Scholarship 2023, 2024, 2025: Recognized as one of the outstanding graduate student working in the field of Electronics Packaging. **(Dec 2023, 2024, 2025)**

IEEE ITherm 2024 Best Poster Runner-Up Award: Awarded among 50+ presenters, in Emerging Technologies and Fundamentals Track. **(May 2024)**

ASME Interpack 2023 Top 10 Best Poster Award: Awarded with the best poster award after being selected among the top ten best poster presenters. **(Oct 2023)**

Full Governmental Scholarship: Awarded a full governmental scholarship on merit for studying bachelor's in mechanical engineering at Tribhuvan University, Institute of Engineering, Dharan, Nepal. **(Sept 2014)**

Publications and Conference Presentations

Selected Publications:

S. Bimali, R. V. Kulkarni and A. M. Marconnet, "Design of a Metrology Tool to Evaluate Degradation of Thermal Greases with Dynamic Warpage," 2025 25th IEEE Intersociety Conference on Thermal and Thermomechanical Phenomena in Electronic Systems (ITherm), Orlando, FL, USA, 2026

Lall, P., **Bimali, S.**, and Miller, S. (August 17, 2024). "Process Recipe and Functional Circuitry Performance on Aerosol Jet Printed Water-Based Silver Ink." ASME. J. Electron. Package. December 2024

P. Lall, **S. Bimali**, V. Soni and S. Miller, "Process Recipes of Additively Printed Sustainable Silver-Ink Using Aerosol Jet Printing," 2023 22nd IEEE Intersociety Conference on Thermal and Thermomechanical Phenomena in Electronic Systems (ITherm), Orlando, FL, USA, 2023

P. Lall, **S. Bimali** and S. Miller, "Advancing Sustainability in Printed Electronics: Low-Temperature Interconnects and Water-Based Ink Performance," 2024 23rd IEEE Intersociety Conference on Thermal and Thermomechanical Phenomena in Electronic Systems (ITherm), Aurora, CO, USA, 2024

Conference Presentations:

S. Bimali, A.M. Marconnet, "Design of a Metrology Tool to Evaluate Degradation of Thermal Greases with Dynamic Warpage", *ITherm*, Orlando, FL, 26 - 29 May, 2026.

S. Bimali, P. Lall,, "Advancing Sustainability in Printed Electronics: Low-Temperature Interconnects and Water-Based Ink Performance", *ITherm*, Denver, CO, 28 - 31 May, 2024.

S. Bimali, P. Lall, J. Narangaparambil, & S. Miller. "Component attach process recipe and performance on aerosol printed sustainable silver ink", *ASME INTERPACK*, San Diego, California, USA. October 24–26, 2023

S. Bimali, P. Lall, V. Soni, & S. Miller,"Process Recipes of Additively Printed Sustainable Silver-Ink Using Aerosol Jet Printing", *ITherm*, Orlando, FL, 30 May - 2 June, 2023.